Suspicious Activity Classification Using CNN and Transfer Learning

Machine Learning Project Phase II – Proposal

Student Name

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Course

Machine Learning

Project Title

Suspicious Activity Classification Using CNN and Transfer Learning

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1. Background and Motivation

Public safety and security are important concerns in modern society. Surveillance cameras are widely deployed in public places such as shopping malls, transportation hubs, schools, offices, and streets. However, monitoring large volumes of surveillance footage manually is time-consuming and prone to human error.

Recent advances in Artificial Intelligence (AI), Machine Learning (ML), and Computer Vision have enabled automated analysis of surveillance images and videos. Convolutional Neural Networks (CNNs) have shown remarkable performance in image classification tasks and can assist security personnel by automatically detecting suspicious activities.

This project aims to develop a deep learning-based image classification system capable of distinguishing between violent and non-violent activities using surveillance images.

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2. Problem Statement

Manual surveillance monitoring is inefficient when large amounts of video data are generated continuously. Security personnel may miss important incidents due to fatigue, distraction, or information overload.

The objective of this project is to develop a CNN-based classification model capable of automatically identifying suspicious activities by classifying images into:

- Violence
- Non-Violence

The developed model can support intelligent surveillance systems by providing automated alerts for potentially dangerous situations.

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3. Dataset Information

Dataset Name

Real Life Violence Situations Dataset

Dataset Source

Kaggle

Dataset Link

https://www.kaggle.com/datasets/abdulmananraja/real-life-violence-situations

Dataset Description

The dataset contains images representing violent and non-violent situations collected from real-world scenarios. The images are categorized into two classes and are suitable for binary image classification tasks.

Number of Images
Violence Images: 5,842

Non-Violence Images: 5,231
Total Images: 11,073

Number of Classes

2

1. Violence
2. Non-Violence

Image Type

RGB Images

Image Size

Resized to 224 × 224 pixels during preprocessing.

Class Distribution

Class	Approximate Images
Violence	5,842
Non-Violence	5,231

Total Images: Approximately 11,000+

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4. Research Questions

1. Can CNN models effectively classify suspicious activities from surveillance images?
2. How does a custom CNN compare with transfer learning models for violence detection?
3. Which model achieves the highest classification accuracy on the selected dataset?
4. Can transfer learning improve model generalization and reduce overfitting?

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5. Project Objectives

The objectives of this project are:

- To develop a custom CNN model for suspicious activity classification.
- To implement transfer learning models using ResNet50 and MobileNetV2.
- To compare the performance of different deep learning architectures.
- To evaluate model performance using multiple classification metrics.
- To identify the most effective model for violence detection.

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6. Proposed Methodology

The project follows the following workflow:

Step 1: Data Collection

Obtain the Real Life Violence Situations Dataset from Kaggle.

Step 2: Data Preprocessing

- Image resizing
- Image normalization
- Dataset loading
- Data augmentation

Step 3: Data Augmentation

The following augmentation techniques will be applied:

- Horizontal flipping
- Rotation
- Zooming

These techniques increase dataset diversity and reduce overfitting.

Step 4: CNN Development

A custom Convolutional Neural Network will be developed using:

- Convolution Layers
- Max Pooling Layers
- Dense Layers
- Dropout Layers

Step 5: Transfer Learning

Two transfer learning models will be implemented:

- ResNet50
- MobileNetV2

Pre-trained ImageNet weights will be utilized.

Step 6: Model Training

Models will be trained using:

- Binary Cross-Entropy Loss
- Adam Optimizer

Step 7: Model Evaluation

Performance will be evaluated using:

- Accuracy
- Precision
- Recall
- F1-Score
- Confusion Matrix
- Classification Report

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7. CNN Model Design

The custom CNN architecture consists of:

1. Input Layer (224 × 224 × 3)
2. Convolution Layer (32 Filters)
3. Max Pooling Layer
4. Convolution Layer (64 Filters)
5. Max Pooling Layer
6. Convolution Layer (128 Filters)
7. Max Pooling Layer
8. Flatten Layer
9. Dense Layer
10. Dropout Layer
11. Output Layer (Sigmoid Activation)

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8. Transfer Learning Models

ResNet50

ResNet50 is a deep residual network pre-trained on the ImageNet dataset. It is widely used in image classification tasks due to its ability to learn complex visual features.

MobileNetV2

MobileNetV2 is a lightweight convolutional neural network optimized for efficient image classification. It provides an excellent balance between accuracy and computational efficiency.

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9. Evaluation Metrics

The following metrics will be used:

Accuracy

Measures overall classification performance.

Precision

Measures how many predicted positive samples are actually positive.

Recall

Measures how many actual positive samples are correctly identified.

F1-Score

Provides a balanced measure between precision and recall.

Confusion Matrix

Displays correct and incorrect classifications.

Classification Report

Summarizes precision, recall, F1-score, and support values.

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10. Experimental Results

The following results were obtained after training and evaluation.

Custom CNN

Metric	Value
Training Accuracy	88.47%
Validation Accuracy	73.49%

ResNet50

Metric	Value
Training Accuracy	63.28%
Validation Accuracy	54.07%

MobileNetV2

Metric	Value
Training Accuracy	92.63%

Metric	Value
Validation Accuracy	88.03%

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11. Classification Performance of MobileNetV2

Class	Precision	Recall	F1-Score
Non-Violence	0.88	0.87	0.88
Violence	0.89	0.89	0.89

Overall Accuracy: 88%

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12. Model Comparison

Model	Training Accuracy	Validation Accuracy
Custom CNN	88.47%	73.49%
ResNet50	63.28%	54.07%
MobileNetV2	92.63%	88.03%

Among the evaluated models, MobileNetV2 achieved the highest validation accuracy of 88.03%, demonstrating superior performance compared to the custom CNN and ResNet50 models.

13. Expected Figures and Tables

The final report will include:

- Sample Dataset Images
- Accuracy Curves
- Loss Curves
- Confusion Matrix
- Classification Report
- Model Comparison Table

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14. Conclusion

This project proposes a CNN-based suspicious activity classification system using surveillance images. Three deep learning models were evaluated: a custom CNN, ResNet50, and MobileNetV2.

Experimental results demonstrated that MobileNetV2 achieved the highest validation accuracy of 88.03%, outperforming both the custom CNN and ResNet50 models. The model also achieved strong precision, recall, and F1-scores, indicating effective classification of violent and non-violent activities.

The proposed system demonstrates the potential of deep learning techniques for intelligent surveillance and automated suspicious activity detection.